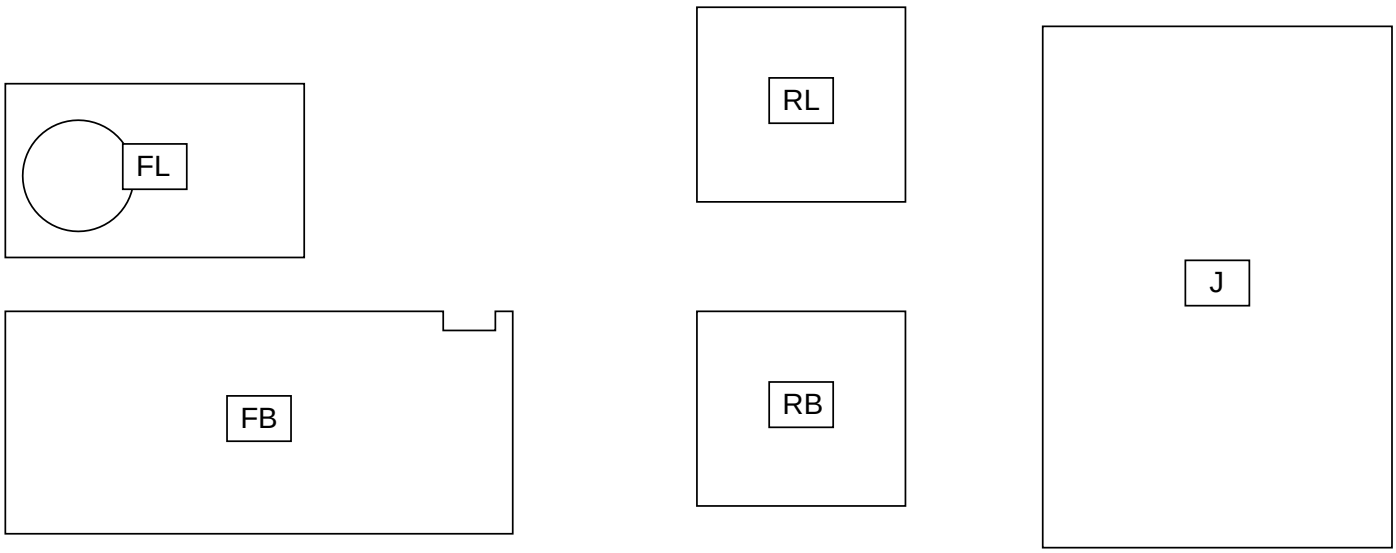
Identify five parts and the trial hardware

This packet covers ONE five-piece coupon plate and ONE metal joint.
FB front-base; FL front-lid; RB rear-base; RL rear-lid; J rear-jig.
One M3 x 8 socket-head screw and one nominal RX-M3x5.7 insert.
Printing tests the catalogue profile; actual insert stock is unconfirmed.
After printing, check label/OD/length BEFORE heat-setting (page 2).

Exact bed sections at0.5 mm; +Y up / +X right

Dimensions in mm; mark part codes before removing the plate layout.



Keep this orientation while removing parts. Mark FB / FL / RB / RL / J
on noncontact outside faces with a fine marker; keep lands and slots clean.
FB/FL: screw corner and guide. RB/RL: hook and pocket. J: tall stop wall.
Base floors and jig floor print down; lid exterior faces print down.

Tools and handling

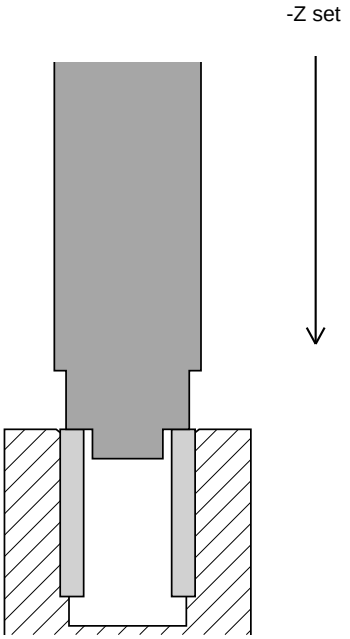
Caliper/depth gauge (tip <=1 mm), controlled M3 insert heater, socket key
matching the real screw (shaft <=3 mm), ONE padded bench clamp reused
front to rear: upper pad <=8 wide x5 deep x5 high; opening >= bench T+4.
Use a 0-10 N gauge/grip (pages3-4), timer, bench and mat.
Keep the bench under the complete base/jig floor. Actual tools must clear
every part. Mark only noncontact outer faces; keep lands and slots clean.
Cool parts, remove loose strings/swarf; reject cracks or sagged lands.
Record caliper resolution for the 0.1 mm stop and 0.2 mm gap readings.
Do not drill out a trial fit or force a part to hide interference.

YOKE / R4 COUPON TRIAL / 2 OF 4

Check stock, set one insert, then slot together

1. Measure actual hardware before applying heat. Match RX-M3x5.7 label:
   catalogue OD4.6 x length5.7, lead-in3.9, pilot4.0 mm; brass RX profile.
   If stock differs, STOP heat-setting and record the measurements.

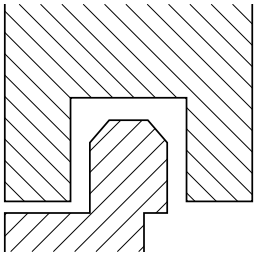
A-A: FB + insert + engaged heater; Y1.6 / X-Z



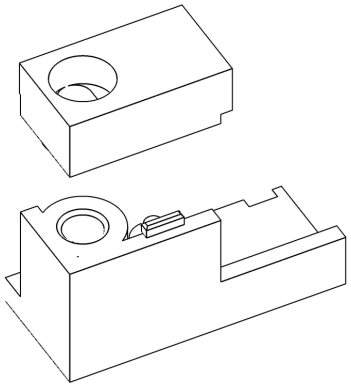
2. Support FB floor flat, with its pilot upright.
   Use maker heat-setting instructions for the
   actual plastic/insert. No temperature is assumed.
   Seat ONE insert coaxially and flush with FB's
   Z13.5 boss face. Let it cool fully before loading.
Tool envelope: OD2.4 pilot x1; OD4.2 shoulder
x2 above it; OD5 body. Pilot is shown 1 mm
engaged. Actual heater must fit that envelope.
Nominal pilot depth6.7 is our length+1.0 allowance.
Boss wall1.9 exceeds catalogue minimum1.6 mm.
Require no crack, whitening, tilt or loose insert.
Hatched: plastic. Gray: nominal metal/tool.

3. With NO screw fitted, hold FL level at X=-1.2 mm from aligned faces.
   Lower over FB's guide; slide RIGHT +1.2 mm until outside faces align.
   Require easy hand seating and real guide restraint, without forced flex.

B-B: actual front guide at X1.5; Y-Z



1.0 tongue /1.5 slot /0.85 walls (mm)



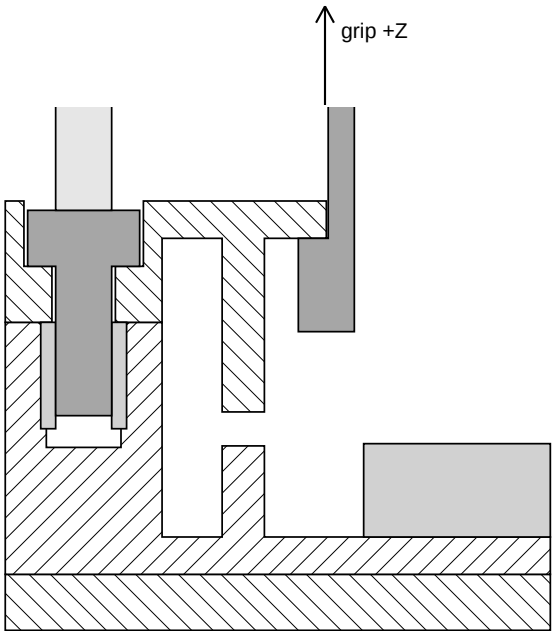
FL above / FB below; exact cropped meshes

COUPON TRIAL ONLY / actual stock and fit unconfirmed
All physical measurements and tests: NOT RUN

Clamp the front floor; keep the screw fitted

C-C at Y1.6: actual screw/driver and clamp ledge in X-Z

Front grip at Y0 is projected; gray shapes are tool envelopes.



Bench fully supports FB; clamp body outside X20.

4. Measure before tightening; nominal mm:
head land16.5 / insert top13.5 => G=3.0
screw L=8 => axial overlap L-G=5.0
insert-end reserve5.7-(L-G)=0.70
plastic reserve6.7-(L-G)=1.70
well3.5 deep / head<=3 => recess0.50.
Require measured overlap>=4.5,
insert-end reserve>=0.4 and recess>=0.2.
Smooth envelopes do not prove usable threads: record thread-start/seat turns.
Head -> FL boss -> FB stop -> insert.
5. Fit ONE M3 x8 only after free seating. Start by hand and lightly seat. Reject bottoming, binding or insert rotation.

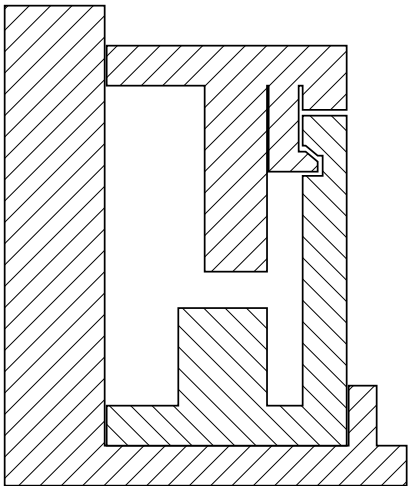
6. KEEP SCREW FITTED. Put FB's complete floor on the bench. Clamp its exposed ledge, X10..20/Y-0.5..7.5, from +X; upper jaw height<=5 mm. The reused small pad fits this maximum 10-deep x8-wide envelope. Open beyond bench T+4; enter with jaw +0.2 Z, then lower 0.2 to seat.
7. Grip FL at X8/Y0. Approach from +X, 0.2 mm below its roof, then raise 0.2 to contact. Hook intrusion under roof<=1.5; below-roof reach<=5 mm. Pull +Z to 5 N for 10 s, three trials. Do not pull on FB or the clamp. Require no release, crack, whitening, insert motion or residual gap>0.2.
8. Remove screw, slide FL LEFT -1.2, then lift. Repeat assembly/service ten times. Record actual turns, dimensions, loads, gaps and damage. No torque is qualified. Stop for forced fit or any failed limit.

YOKE / R4 COUPON TRIAL / 4 OF 4

Hold RB down; leave RL free for the pull

9. OUTSIDE J, lower RL at -1.2 X relative to RB; slide +1.2 to align.
   Lower the CLOSED pair into J; RB rests on J's 2 mm floor.

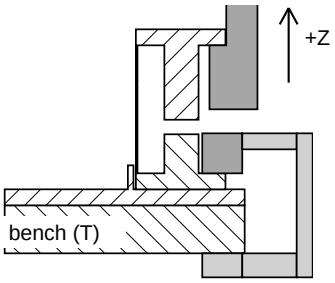
D-D at Y3: exact RB / RL / J; X-Z



J floor / RB middle / RL upper. Stop wall5 mm.

Hard X faces55.5 /67.7; nominal end gaps0.1 each (0.2 reverse travel).
10. Hand-hold J FLOOR for the unloaded free-pair check: vertical drag<0.5 N.
11. THEN clamp RB+J to the bench through the rear window as E-E below. RL is free.
Upper pad<=8 wide x5 deep x5 high, X55.9..63.9/Y6.2..11.2/Z4..9.
Require RB stays seated on J; lifting RB invalidates the fixture, not the joint.
12. Grip RL at X57.2/Y9.1 from +Y:
0.2 below roof, then+0.2 to contact.
Grip width<=2; below-roof reach<=8 mm.

E-E at X60.55: floors/jaw; grip at X57.2 projected (Y-Z)



Upper pad touches ONLY RB's floor.
J lies between RB and the bench.
Open beyond T+4; enter from +Y with upper pad +0.2 Z, then lower 0.2 to seat.
Lower jaw below bench; frame outside the pair. Entire J floor is supported.
T=actual bench thickness. C-frame is schematic: verify actual tool clearance.
Never clamp RL. Dark grip is projected.

13. Pull RL +Z to 5 N for 10 s, three trials. Require stop deflection<=0.1 and relative reverse X travel<=0.3 mm under load. Greater motion or RB lifting INVALIDATES THE FIXTURE; repair it before judging the hook. Require no release, crack, whitening or permanent distortion. Record loaded gap; require residual gap<=0.2 mm after unloading.
14. Remove the clamp, lift the CLOSED pair out of J, then slide RL -1.2 X before lifting it from RB. Repeat assembly/service ten times; inspect.

STOP on any failed limit. Record actual tool fit, forces and observations.
This trial answers joint fit/retention; it does not establish long-term service strength or usable thread engagement. All physical tests NOT RUN.